

Sultanate of Oman
Nama Water Services Company SAOC

**Consultancy Services for Design and Supervision for STP, Sewer and TE Networks
Systems in Ibri**

Tender No. T/2719110/2025

Design Criteria

Sewer Network Design, Iteration W13

Revision 0

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Document control

Revision	Date	Status	Basis
0	7 September 2026	For design use	Project design-criteria register, with the 12 m cover decision and the wadi rules checked against the guideline pages on 7 September 2026; values applied by the W13 design pipeline read from its criteria module on the same date

This document is rebuilt from source on every revision. A revision that has been issued is not overwritten; the next revision is written to its own folder.

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1 Purpose, sources and how to read this document

This document sets out the design criteria for the Ibri sewer network. Every numeric value is taken from one of the three Nama Water Services design guidelines and cited to its page, or is a stated project decision or assumption recorded in Sections 20 and 21. No value has been taken from memory or from general practice.

1.1 Sources

Code	Document	What it governs here
G203	PAM-GUD-203 Wastewater Design Guidelines, 201 pages	Gravity sewers, chambers, pumping stations, force mains, treatment plants, odour, surveys
G1	PAM-GUD-201 General Design Guidelines, 152 pages	Population, demand and flow estimation, design life, options appraisal, wadi and road crossings, modelling
G2	PAM-GUD-202 Water and TSE Design Guidelines v1.0, 177 pages	Treated effluent and water networks
Scope	Terms of Reference, Tender T/2719110/2025	Design horizons, model years, deliverables, inverted siphons
R0	Inception Report Revision 0 and its demand workbook, August 2026	Project values already adopted, reconciled against the guideline in Appendix A

1.2 Citation convention

A reference such as G203-p33 §4.6.3 means page 33, clause 4.6.3 of PAM-GUD-203. G1 and G2 refer to PAM-GUD-201 and PAM-GUD-202 in the same way. Page numbers are the printed page numbers of the guideline PDF.

Where the guideline's wording decides how a rule applies, the phrase is quoted. Shall and must are requirements. Recommended and should are guidance the design follows unless a stated reason exists not to. A value marked as a project decision or a project assumption is ours, not the guideline's, and is listed in the registers at the end.

1.3 Precedence

An NWS-approved demand calculation, if one exists, outranks the whole flow-estimation chain in Section 12. The domestic consumption values in the guideline are indicative and are to be validated by NWS before design. Where the guideline gives two methods, the one written as a requirement governs and the other is a labelled fallback; Section 12.1 sets this out.¹

1.4 The W13 design and the column "Adopted in W13"

The network design is developed in iterations. Iteration W13 is the current design. It is calibrated on a 5.51 km² test area in Ibri town, where it produces 1,415 chambers, 71.6 km of sewer, a deepest cover of 10.45 m and no pumping station. The design must reproduce that result after every change before it is extended to the 531 km² study area.²

Where a criterion is implemented in the design pipeline, the value applied is shown in the column "Adopted in W13". A blank cell means the criterion applies at a later stage or to a system the pipeline does not yet design. Sections 15 to 19 concern the treatment plant, the treated effluent system, odour, modelling and surveys, and carry no such column.

¹ G1-p59 §7.3 and G1-p70 §7.4: "In the absence of a developer-provided water demand calculation based on an approved methodology, the water demand shall be calculated in accordance with the methodology detailed below." G1-p60 Table 11: values "should be validated by NWS as essential design criteria, before designing the project".

² Test-area result of the W13 pipeline on 7 September 2026, identical to the W8 design it was copied from. The built Nama network on the same ground also needs no pumping station.

1.5 Vocabulary

The guideline names the tertiary pipes property connection sewer, rider sewer and lateral sewer. The network hierarchy in the design uses Nama's as-built vocabulary of trunk main, sub main and lateral, read from the manhole identifiers of the built network. Where this document says lateral sewer it means the guideline's tertiary pipe; where it says lateral tier it means the lowest tier of the secondary network.

2 Gravity sewer hydraulics

Table 1 Hydraulic design criteria for gravity sewers (G203 §4.2, pages 24 to 28)

Item	Guideline requirement or value	Reference	Adopted in W13
Design method	Colebrook-White or Manning; licensed software approved by NWS. The scope names SewerGEMS	G203-p24	Colebrook-White on the true internal bore; SewerGEMS as the referee model
Roughness, Colebrook-White	ks = 1.5 mm , all pipe sizes and materials	G203-p24, p28	1.5 mm
Kinematic viscosity	$1.141 \times 10^{-6} \text{ m}^2/\text{s}$ at 15 °C, conservative basic design	G203-p25	$1.141 \times 10^{-6} \text{ m}^2/\text{s}$
Self-cleansing velocity	0.75 m/s at peak flow ; 0.90 m/s preferred	G203-p26	0.75 m/s required at every pipe; 0.90 m/s reported
Maximum velocity	3.0 m/s at design depth of flow	G203-p27, p29	3.0 m/s
Proportional depth at peak flow	$d/D \leq 0.65$ for $D \leq 350 \text{ mm}$; $d/D \leq 0.50$ for $D > 350 \text{ mm}$	G203-p27 Table 10	As guideline, threshold at 350 mm
Manning n	PVC and GRP 0.009 to 0.011; PE 0.009 to 0.015; cement-lined concrete 0.012	G203-p23 Table 8	0.013 as the ks-equivalent for the SewerGEMS export
Minimum gradient method	"Steeper gradient calculated based on self-cleansing velocity and minimum tractive force methodology shall be adopted as minimum pipe gradient." The tractive-force method applies at network heads where 0.75 m/s cannot be reached	G203-p27 §4.2.2.1	The steeper of Table 11 (Section 3) and Equation 1
Early-phase low flows	Flow in the early development phases is below the design flow, with a clogging risk at low velocity; more frequent inspection and cleansing in that period	G203-p28 §4.2.6	Self-cleansing verified at start-year flows; see Section 12.3 on the direction of accuracy

2.1 Minimum gradient by tractive force

The guideline gives the tractive-force method of Mara, Sleigh and Taylor for the minimum gradient, with a proportional depth of 0.2 and a Manning n of 0.013 behind it. It gives no numeric design value of the tractive tension; that value is a project assumption to be confirmed with NWS.¹

$$S_{\min} = K \cdot \tau^{1.23} \cdot Q^{-0.461} \quad (1)$$

Symbol	Meaning	Unit
S _{min}	Minimum gradient	m/m
K	2.33×10^{-4} with Q in m ³ /s; 5.5×10^{-3} with Q in L/s	–
τ	Tractive tension; project assumption 1 Pa (Section 20)	Pa
Q	Design flow, floored at 1.5 L/s, Mara's minimum design flow (Section 20)	m ³ /s

At the 1.5 L/s floor the tractive-force gradient is close to the Table 11 value for DN200, so the two methods meet at the network head. If NWS sets the tension at 2 Pa rather than 1 Pa, a large share of the smallest pipes need a steeper gradient; the sensitivity is run by substituting the value, not by editing the design.

¹ G203-p27 §4.2.2.1. The equation is printed in the guideline as an image. Its simplified form without the tension term is valid only at τ = 1 Pa.

3 Minimum and maximum gradients

Table 11 of the guideline gives the minimum gradient of the secondary network by diameter, computed with Colebrook-White at 0.75 m/s. The design pipeline reproduces these values from the Colebrook-White equation within 5 % before any design runs, as a check on the hydraulic code.¹

Table 2 Minimum gradients of the secondary network (G203-p29 Table 11)

Nominal diameter (mm)	Minimum gradient (mm/m)	Minimum gradient (%)
200	5.00	0.500
250	3.75	0.375
315	2.70	0.270
400	2.05	0.205
500	1.55	0.155
600	1.25	0.125
700	1.00	0.100
800	0.85	0.085
900 and larger	0.75	0.075

Table 3 Rules on gradient (G203-p29 §4.3.1; G203-p18 Table 5)

Item	Guideline requirement or value	Reference	Adopted in W13
Oversizing	No oversizing of a pipe to obtain a flatter slope; uniform slope between chambers	G203-p29	Diameter chosen for the flow, never for the slope
Maximum gradient	Governed by the maximum velocity of 3.0 m/s	G203-p29	3.0 m/s
Construction tolerance	"The lines and level of any pipeline shall not deviate from that described in the contract by more than 20 mm and combination of such deviation shall not create a reverse gradient." At DN900 and above the minimum gradient is 0.75 mm/m, so 20 mm over a 120 m spacing is about 22 % of the available fall	G203-p29 §4.3.1	A fall tolerance of 0.040 m, twice the 20 mm, is carried in the invert solver so a flat profile keeps its margin
Round gradient values	–	Project decision	Pipes are laid at round 0.05 % steps so the value on the drawing is the value the invert levels came from. A single pipe is rounded up; a whole run is eased down to the next round value, which leaves the far end slightly shallower and never deeper. Measured cost on the test area: 1.0 % more excavation, 0.12 m on the deepest chamber, no additional pumping station
Tertiary slopes	Property connection sewer 3 % to 10 % ; rider sewer 1 % to 10 % ; lateral sewer 1 % to 10 % . Table 11 applies to the secondary network only; its 0.5 % at DN200 is not a lateral-sewer value	G203-p18 Table 5	3 % minimum for property connections, 1 % for riders and lateral sewers, 10 % maximum

¹ G203-p29 §4.3.1 Table 11.

4 Pipes

Table 4 Pipe diameters and materials (G203 §4.1 pages 21 to 23; §5 page 35)

Item	Guideline requirement or value	Reference	Adopted in W13
Minimum diameters	Property connection OD160; lateral sewer OD200 with a maximum length of 45 m; main sewer OD200 minimum	G203-p22 Table 6	OD160 tertiary; DN200 minimum main; diameter series 200, 250, 315, 400, 500, 600, 700, 800, 900, 1000, 1200
Secondary network range	200 to 400 mm typical; 400 mm is not a mandatory ceiling	G203-p23	
Materials, main sewer 350 mm and larger	GRP, HDPE or lined reinforced concrete in open trench; GRP or HDPE for trenchless installation	G203-p22	PVC-U through the OD series to 315 mm; GRP above
PVC-U size series	OD-designated series to 315 mm	G203-p23 Table 7	As guideline
Trunk main definition	Diameter over 800 mm, length over 1,000 m without connections, upstream of the treatment plant or a main pumping station	G203-p35	
Trunk material above 600 mm	GRP, lined reinforced concrete, profile-wall HDPE	G203-p35 Table 14	
Internal bore for hydraulics	Plastic mains are OD-designated, so the bore is smaller than the nominal size	G203-p22 Table 6	PVC-U taken as SDR34 (SN8): $\text{bore} = \text{OD} \times (1 - 2/34)$. GRP nominal size is the bore. The wall class is pending PAM-SPC-207 (Section 20)
Crown-to-invert height	–	Project assumption	Outside diameter used for the geometry, nominal for GRP; a 0.05 m allowance for wall and bedding below the crown cover

5 Depth, cover and corridors

Table 5 Cover and clearance (G203 §3.5 page 19; §4.6 pages 32 and 33)

Item	Guideline requirement or value	Reference	Adopted in W13
Minimum cover, gravity sewer	"The minimum depth for sewer pipes shall be 1.3 m to the crown"	G203-p33 §4.6.3	1.3 m to crown
Reduced cover	"If circumstances require installation of a pipe with depth less than 1.3 m above the crown, then concrete protection is required. The minimum cover above the pipe and its protection shall be 0.5 m." The exception needs a stated circumstance, concrete protection, and the 0.5 m measured above the protection	G203-p33 §4.6.3	Not used
Property connection cover	600 mm minimum	G203-p19	0.60 m
Maximum cover	"The recommended maximum cover for sewer pipes is approximately 10 – 12 m. Depths with cover greater than this shall be investigated with pipe manufacturers ... Where the cost of excavation becomes prohibitive the Engineer shall incorporate pumping stations into the design."	G203-p33 §4.6.3	12.0 m of cover as a limit with no exception , checked at every chamber and along the trench between chambers (Section 5.1)
Horizontal clearance to other utilities	3 m	G203-p33	
Service corridor widths	DN200 to 500: 2.0 m; 600 to 900: 2.8 m; 1000 to 1200: 3.2 m; 1400 to 1700: 4.0 m; 1800: 4.1 m; 2000 to 2400: 4.4 m	G203-p32 Table 13; p35 Table 15	
Position in the carriageway	At least 1 m from the kerb line, stated for force mains	G203-p51	Applied to gravity sewers by inference; chambers at least 0.5 m from the kerb (Section 20)

5.1 The 12 m limit

The guideline makes the cost of excavation, not the depth itself, the trigger for a pumping station, and treats 10 to 12 m of cover as a recommendation to be checked with pipe manufacturers. No cost analysis exists at concept stage, so the design applies 12 m of cover as a limit with no exception. Where a gravity sewer would pass 12 m, a pumping station is placed before that point, the sewage is lifted, and the sewer restarts at normal cover.¹

The limit is revisited only when the cost analysis exists, and then for each excursion on its own economics, not as a blanket relaxation. Pumping stations are not removed by digging deeper: the route is searched again first, and whatever pumping remains is real.

¹ Project decision of 7 September 2026. The limit binds nowhere on the test area, whose deepest cover is 10.45 m; it will bind when the design is extended to the whole study area, and that is where the cost question becomes real.

6 Chambers

Table 6 Chamber criteria (G203 §4.4 pages 29 to 31)

Item	Guideline requirement or value	Reference	Adopted in W13
Maximum spacing	DN200 to 315: 100 m ; DN350 to 900: 120 m ; DN1000 to 1400: 150 m ; above DN1400: 200 m . Deviation needs NWS pre-approval	G203-p30 Table 12	Runs split at 100 m, which satisfies every class; spacing rounded to 10 m, with 5 m as the fallback
Locations	Change of gradient or diameter, junctions, end of laterals, and at the regular spacing	G203-p29	As guideline
Backdrop	Required where the invert drop exceeds 600 mm; external; maximum height 2 m, beyond which a vortex drop shaft; internal only in a chamber of 1.5 m diameter or more	G203-p30	Drop trigger 0.60 m; backdrop maximum 2.0 m
Inlet angle	At least 90° to the direction of flow ("shall")	G203-p30	85° minimum applied ; an inlet sharper than 75° is flagged for a purpose-made chamber with a curved channel and is never fixed by adding a chamber. Stated deviation, Section 21
Minimum separation of chambers	None stated in any of the three guidelines	–	3.0 m as a physical convention; two chambers closer than 3 m become one structure
Chamber size	"Sufficient size"; no table given	G203 §4.4	DN1000 to 3 m depth, DN1200 to 6 m, DN1500 deeper; no hydraulic effect at concept
Prohibited locations	Wadis and flood-prone or washout areas, for pipelines and chambers alike	G203-p30 §4.4.1; p33	Flood hazard classes 4, 5 and 6 excluded (Section 8)
Outlets per structure	–	Project convention	One physical outlet per chamber. A branch leaving a chamber that already has an outlet starts at the next house connection, or 10 m away
Clearance from plots	–	Project decision	Every chamber is checked against every plot boundary and slid clear; a corner chamber must be 2 m clear of any plot (Section 9)

7 Property connections, riders and lateral sewers

The tertiary layer connects each property to the network. At concept stage its only hydraulic test is whether the property can drain by gravity to the sewer in its street; the pipes are not sized.

Table 7 Tertiary criteria (G203 §3, pages 17 to 22)

Item	Guideline requirement or value	Reference	Adopted in W13
Property connection sewer	OD160; slope 3 % to 10 %; cover 600 mm minimum; maximum length 50 m	G203-p22 Table 6; p18 Table 5; p19; p18 Table 4 note	As guideline
Rider sewer	Slope 1 % to 10 %. "Several HCC (usually up to 3) may be connected together by one or several Rider Sewers within the public ROW."	G203-p18 Table 5; p19 §3.4	Up to 3 house connections per rider; the rider runs along the frontage in the public right of way; a lone house connects by a lateral sewer without a rider
Lateral sewer	OD200; slope 1 % to 10 %; maximum length 45 m. Table 6 states the 45 m on the lateral sewer row only, while §3.2 attaches it to rider and lateral alike	G203-p22 Table 6; p17 §3.2	45 m applied to riders and lateral sewers, the conservative reading, as a declared project cap
Connection point	–	Project decision	The perpendicular projection of the plot onto the corridor it fronts, never the plot centroid; the connection is a short perpendicular spur; the plot loads the reach it fronts
Gravity viability check	–	Project assumption	The house drain leaves the plot 0.60 m below plot ground and is checked against the pipe invert interpolated at the connection chainage, at a blended fall of 2 %
Stub-outs	–	Project doctrine	A capped connection at the frontage of every planned plot, sized for its saturation flow; DN200 minimum usually governs
Output layers	–	Project decision	House connections and riders sit in their own layers in every output, never in the mains layer

8 Wadis, flood hazard and crossings

Table 8 Wadi rules (G203 pages 30, 33 and 52; G203 §11.5.3.1)

Item	Guideline requirement or value	Reference	Adopted in W13
Pipelines and chambers in wadis	"Locating pipelines and associated chambers in wadis or areas subject to washout during heavy storms must be avoided." Page 33 repeats it as "shall be avoided"	G203-p30 §4.4.1(a); p33	The design avoids it. Where no other route exists, the presence is a recorded and justified exception carried to the deliverable, never a silent one (project decision, 7 September 2026)
What counts as wadi ground	Areas subject to washout during heavy storms	G203-p30	Classes 4, 5 and 6 of the 50-year flood hazard grid, which are flood-hazard classes keyed on danger to people and vehicles standing in for the washout criterion; classes 1 to 3 and cells with no value are dry. Project assumption, to be settled by a scour-depth check (Section 20)
Cover at a wadi crossing	1.5 m to crown , gravity sewer and force main alike. The clause opens "As for gravitational sewer, the minimum cover should be..." and lists 1.3 m unprotected, 0.5 m protected and 1.5 m at a wadi crossing	G203-p52 §8.2.4	1.5 m
Reduced cover at a crossing	Whether the 0.5 m protected exception is available at a wadi crossing is not stated	G203-p33; p52	Not available: the 1.5 m answers scour and concrete encasement answers external load. Project position, recorded as ours
Twin pipelines at an obstacle crossing	Allowed with a dedicated hydraulic justification: 0.75 m/s in all modes, mechanically independent restraint	G203-p52 §8.2.3	
Inverted siphons	"Inverted siphons shall not be allowed." The scope says avoid, and only where no other feasible means exists	G203-p182 §11.5.3.1(c); scope p12(60)	None. Any siphon is a formal deviation needing justification
Trenchless at major crossings	An alternative subject to NWS approval; settlement is decisive	G203-p21, p35	Every dual-carriageway crossing is assumed trenchless (Section 9)
Treated effluent discharge to a wadi	Class A at least under MD 145/93; Environmental Authority and APSR approval	G203-p73	

8.1 Crossing procedure

Running along a wadi and crossing one are different questions. The general guideline sets the procedure for a crossing.

Table 9 Wadi, road and falaj crossings (G1 §9.3, pages 85 and 86)

Item	Requirement or value	Reference
Data and approvals	Wadi bed profiles and cross-sections, flood frequency at 1 in 20, 1 in 50 and 1 in 100 years, bed material and bed-level change, from the Civil Aviation Authority and the Ministry of Agriculture, Fisheries and Water Resources; MoAFWR approval required	G1-p85
Pipe material	Ductile iron over the crossing length plus 15 m each side, with mechanical or detachable joints	G1-p86

Item	Requirement or value	Reference
Protection	NWS standard drawing PAM-STD-404; anti-flotation check with the pipe empty under flood or high groundwater	G1-p86
Cover in soft soil	2.0 m minimum	G1-p86
Valves	Isolation and air valves on both sides of an active or major crossing; washout at the low point on one side; no chambers or markers in the wadi bed or embankments; everything accessible during a flood	G1-p86
Road crossings	Trenchless preferred; reinstatement to the Oman Highway Design Manual	G1-p85
Falaj crossings	Buffer zones, protection and minimum safe excavation distances; no-objection certificates sought at the preliminary stage	G1-p86; G203-p39

9 Layout rules for the network

The guidelines fix the hydraulics and the cover. They say little about where a sewer may run through a town. The rules below were set for this project from the road data supplied and from the shape of Nama's built network, and each states its basis.¹

Table 10 Layout rules and their basis

Rule	Statement	Basis
Road source	Corridors come from the road centreline layer. The dual attribute decides exclusion: 1 is a dual carriageway, 2 is a two-lane pair of which one side is used. Road class serves preference only, not exclusion: 150 national and arterial roads are single carriageways	Project decision, 19 August 2026
Dual carriageways	No pipe of any kind runs along a dual carriageway, the trunk included, because the carriageway cannot be opened. Both lines of the pair are removed with their links and ramps. A line within 6 m and 25° of an excluded carriageway is treated as its twin and excluded too	Project decision, 19 and 20 August 2026
Crossing a dual carriageway	Only as a short perpendicular pipe: within 25° of square, no longer than 70 m, trenchless, no chamber on the carriageway. The route search is charged 2,500 m of equivalent route for every crossing, and nothing within 30 m of a known underpass, so a crossing is used only where it pays	Project decision, 20 August 2026; trenchless per G1-p85
Two-lane pairs	One side only, held for the whole corridor: the side with more fronting plots, tie-broken by lower ground. Plots on the far side connect by a direct link at the road end	Project decision, 19 August 2026
Roundabouts	Carry no sewer; the approach legs terminate at the ring. A ring is a roundabout only if no plot lies inside it, every node has an approach arm and the arcs are curved, with a perimeter under 150 m and an equivalent radius under 30 m	Project decision, 19 August 2026
Traffic links	Turning fillets, slip roads and diagonal links between carriageways exist for vehicles; the sewer joins at a point. A link is dropped when it serves no plot within 40 m, both ends are attached, the way round is no more than three times its length, and it is short (120 m or less) or strongly curved (45° or more). Dead-ends serving no plot are removed, repeatedly	Project decision, 19 and 20 August 2026
One line per street	A straight street between two intersections is one polyline; a break of less than 10° is dissolved	Project decision
Head of a run	A run starts at the first house gate on the street, taken as the plot centroid dropped square onto the street and searched up to 45 m, not at the street end	Project decision, 20 August 2026
Chambers at bends	Up to 5°: none, the joint deflection absorbs it. 5° to 45°: one chamber at the bend. A sweeping curve over 45°: two or three chambers breaking it into chords, three only on a wide long bend. A corner chamber sits at the intersection of the tangents only if it is 2 m clear of every plot boundary; otherwise the curve is followed. The governing constraint is cleaning access, not pipe flexibility	Project decision, 19 August 2026
Hierarchy	Trunk main, sub main and lateral, in Nama's as-built vocabulary; every pipe carries its tier. Each join onto the main pipe is charged 400 m of equivalent route, so only collectors reach it: 20 joins on the test area, where fewer than 15 costs a pumping station and fewer than 8 starts crossing dual carriageways	Learned from the built network, 23 August 2026
Main pipe	An input drawn by the engineer, not derived: both legs drain to their meeting point and on to the existing treatment plant	Project input, 20 August 2026
Pumping pockets	A pocket of fewer than 50 plots is absorbed at detail design. The rules for cascading stations and for sizing a rising main on pump duty are not settled and will be decided when the design raises them	Project position, 7 September 2026

¹ The built network in the test area, traced through its own manhole identifiers, is one trunk main, six sub mains and 419 laterals; 91 % of laterals drain into another lateral and only about sixteen pipes touch the trunk. The built network runs along a dual carriageway on 0.1 % of its length.

10 Pumping stations

The test area needs no pumping station. The criteria below apply wherever the 12 m limit of Section 5.1 places one in the wider study area.

Table 11 Pumping station criteria (G203 §7, pages 38 to 49)

Item	Requirement or value	Reference
Siting	Hydraulics-driven; flooded suction preferred; the force-main energy-economic study determines the optimal location. The site is approved in advance by NWS at concept or preliminary stage	G203-p38 §7.2
Flood protection	Pump pedestal, building floor, transformers, substation and emergency generator above the maximum flood level; floors at least 300 mm above the 1 in 50 year flood level; surface and stormwater designed for a 1 in 50 year return period; sites under power lines avoided	G203-p38 §7.2
Station type	Type 1 up to 100 L/s, one duty and one standby; Type 2 over 100 to 300 L/s, two duty and one standby; Type 3 over 300 L/s, three duty and one standby. Pumps rotate duty each cycle; station pipework 0.6 to 2.5 m/s (0.5 m/s with a grinder); motor speed at most 1,450 rpm above 5 L/s and 2,800 rpm at or below; solids passage 76 mm minimum, 65 mm with an upstream basket	G203-p40 to 41 Table 17
Small stations	At least two identical pumps, duty and standby, each at 100 % of design flow; peak flow achievable with any one unit out	G203-p39
Land area	Type 1: 50 to 100 m ² ; Type 2: 200 to 400 m ² ; Type 3: 900 m ² or more; access with a turning circle at least 6 m wide and hard standing	G203-p43 Table 21
Design life	Non-structural mechanical installations 20 years; Table 17 rates pumps at 15 years, which is the figure for pump replacement in the life-cycle cost	G203-p38; p40 Table 17
Minimum-flow factors	Initial minimum flow = average flow × factor: 50 L/s → 0.25; 500 L/s → 0.35; 2,500 L/s → 0.45; 5,000 L/s → 0.50. This flow, not the average, sizes the force main against deposition	G203-p40 §7.4 Table 16
Emergency overflow	Every pumping station shall have one, with Environmental Authority approval; never at an upstream manhole; wet-well overflow with dip tube or baffle; storage above the start level of the last duty pump. Prevention by dual power supply, trunk or basin storage, emergency storage and emergency pumping	G203-p46 to 47 §7.5
Wet-well volume	Minimum live volume per Equation 2, with successive start and stop levels 200 to 300 mm apart, a 5 to 10 s start delay, and CFD plus physical modelling for stations of 0.5 m ³ /s or more	G203-p48 §7.8
Net positive suction head	Equation 3, with a margin of at least 1 m	G203-p47 §7.6
Hydrogen sulphide at the force-main outlet	Where H ₂ S is expected and no field data exists, design for management and monitoring at 50 to 100 ppm average and 200 ppm peak at the pressure-main termination; monitoring at the termination manhole controls chemical injection at the force-main start	G203-p47 §7.7; p55 §8.5
Stakeholder approvals	No-objection certificates, including falaj crossings, sought at the preliminary stage	G203-p39

$$V = 0.25 \cdot Q \cdot T \quad (2)$$

Symbol	Meaning	Unit
V	Minimum live volume of the wet well	m ³
Q	Capacity of a single pump	m ³ /s
T	Cycle time, 3600 divided by the starts per hour; at least 10 starts per hour for motors up to 30 kW	s

$$\text{NPSH}_a = H_a - H_{\text{vpa}} - H_{\text{st}} - H_f \quad (3)$$

Symbol	Meaning	Unit
NPSHa	Net positive suction head available; margin of at least 1 m over the pump's requirement	m
Ha	Atmospheric pressure head	m
Hvpa	Vapour pressure head of the liquid	m
Hst	Static suction lift	m
Hf	Friction loss in the suction pipework	m

11 Force mains

Table 12 Force main criteria (G203 §8, pages 50 to 55; G1 Appendix III)

Item	Requirement or value	Reference
Minimum velocity	0.75 m/s continuous, maintained at the design minimum flow of Table 16 under maximum static head; 1.0 m/s intermittent; 1.2 m/s in vertical pipework	G203-p50 §8.1; p40 Table 16
Maximum velocity	2.5 m/s	G203-p50
Minimum diameter	75 mm bore for non-clog pumps; 50 mm bore for grinder pumps	G203-p50 §8.1
Gradients	At least 1 in 500 rising and 1 in 300 falling; never flatter than 1 in 750	G203-p50 §8.2.1
Run full	Designed to run full and remain full at all times	G203-p51 §8.2.1
Retention time	Ideally 30 minutes or less; air valves at high points and washouts at low points	G203-p50
Access	Every 500 m	G203-p50
Valve spacing	In-line isolation valves at about 500 m and never more than 800 m; washout valves at low points sized to empty a section in 3 to 4 hours (main up to 400 mm: 100 mm; 500 to 800 mm: 150 mm; 900 to 1,200 mm: 200 mm; 1,200 mm and larger: 300 mm); double-orifice air valves at high points with the approach no flatter than 1 in 500 and the departure no flatter than 1 in 300, each with its own geared isolation valve	G203-p53 to 54 §8.4
Separation from water mains	3.0 m horizontal; crosses under the water main with 450 mm vertical clearance outside to outside; one full length of water pipe centred so both joints are as far from the force main as possible; structural support where required	G203-p51 §8.2.2
Layout	Straight lines, sharp bends avoided, at least 1 m from the kerb line in a carriageway	G203-p51
Material, in the station	Ductile iron recommended; stainless steel for stations under 100 L/s	G203-p52 §8.3
Material, pressure main	"The recommended pipe material for the pressure main is Ductile Iron and HDPE"	G203-p53 §8.3
Termination	Discharge to the gravity system at a manhole, entering not more than 300 mm above the receiving flow line; water seal and forced venting through odour control where turbulent; corrosion-resistant or lined receiving manhole	G203-p55 §8.5
Surge analysis	Transient analysis in NWS-approved software; simultaneous pump start and stop modelled to N+1; no vapour cavities or column separation; minimum pressure not below the manufacturer's limit or 0.2 bar below atmospheric, whichever is higher; maximum not above the test pressure or the lowest component rating; a zero-roughness run first for the worst case	G1-p146 to 147 Appendix III

12 Flow estimation

12.1 Two methods in the guideline, and which one governs

The general guideline gives two methods for almost every demand and flow parameter. The first is for planning and forecasting across broad service areas. The second is written as a requirement wherever detailed land-use information exists.

Tier	Applies when	Method
A, planning and forecasting	Broad service areas without land-use detail	G1 Table 11 consumption per head with the distributed ratios of 22 % and 14 %; G1 Table 19 return rates of 85 % and 54 %
B, project-specific design	"where detailed land use information is available"	G1 Table 12 unit rates for non-domestic and governmental demand; design flows to BS EN 752 or a stated equivalent, supported by site-specific evidence

Three clauses carry the requirement. Non-domestic consumption "shall be calculated using the reference values presented in the Table 12" where the project provides detailed land use. Governmental consumption "is to be calculated specifically for the project and not as a ratio of domestic consumption". Design flows for project-specific designs "shall be calculated in accordance with a relevant international standard such as BS EN 752", and "calculations shall clearly state which standard has been used".¹

Every value from the planning tier that appears in a deliverable is labelled as a fallback pending the project-specific data, with the data request named. Section 12.3 records the load basis adopted and the reason the G1 Table 12 method cannot yet be applied.

12.2 Population and occupancy

Table 13 Population criteria (G1 §7.2, pages 58 and 59)

Item	Guideline requirement or value	Reference	Adopted in W13
Official source	NCSI population and housing data at governorate, wilayat and settlement level	G1-p58 §7.2.1	NCSI series carried in the Inception Report
Wilayat to settlement	Where forecasts exist only at wilayat level, distribution must follow the latest census settlement shares	G1-p58	As guideline
Below settlement level	Pro rata to the number of electricity accounts, supplied by NWS	G1-p58	33,970 electricity accounts, each counted as a property
Extrapolation limit	NCSI forecasts cover 20 to 25 years. Beyond them a polynomial regression, but "it is not recommended to extrapolate more than ten years beyond the available forecast period"	G1-p58	The 2055 and ultimate cases are defended on land capacity, not on the extrapolated curve
Population from plots	Equation 4. Occupancy rate = population ÷ housing units from NCSI, most recent data, at the geographic scale of the project	G1-p58 to 59 §7.2.2	Properties per plot counted from electricity accounts, 1.46 domestic per matched plot; additional-account entries are separate dwellings

¹ G1-p60 §7.3.2; G1-p61 §7.3.3; G1-p71 §7.4.1. Table 19 describes its own figures as "baseline figures [that] provide a general framework for planning and forecasting purposes across broader service areas" (G1-p70).

Item	Guideline requirement or value	Reference	Adopted in W13
Occupancy rate	Population ÷ housing units	G1-p58	Project basis 5.32 people per domestic property, measured as 2024 settlement population ÷ counted domestic properties over the same 25 settlements. The W13 test-area run carries 5.0 , the value inherited from the design it was copied from; the full-area run is to apply 5.32 (Section 20)
Build-out	"special care must be given in cases of plot subdivision ... development speed must be considered through appropriate phasing assumptions ... development percentages must be applied over the design period ... particularly avoiding overestimation"	G1-p59 note	Pipes and civil works sized on every plot at saturation; dated years use zone totals capped at the zone ceiling (Section 12.3)

$$\text{Population} = N_{\text{plots}} \times n_{\text{prop}} \times \text{OR} \quad (4)$$

Symbol	Meaning	Unit
Nplots	Number of plots in the area	–
nprop	Average properties per plot; counted from electricity accounts, 1.0 only as the fallback for a plot with no account	–
OR	Occupancy rate, people per property	–

The occupancy rate departs from the guideline's definition in one respect. Housing units are not published at settlement level, so properties counted from electricity accounts stand in for them. The figure was validated by coverage: the 25 settlements hold 63.4 % of the wilayat population, and at 5.32 the wilayat implies 34,504 domestic properties against 22,588 held, which is 65.5 %. The two agree within 2.0 points, so 5.32 is a measurement rather than a coverage artefact. It is listed as a deviation in Section 21.¹

12.3 The load basis

The load basis was fixed on 30 August 2026 and is applied without variation. The planning-tier ratios set the volume; land use sets where that volume is placed. The two do not overlap, so the method neither mixes tiers nor double counts.

1. Domestic load = counted domestic accounts × occupancy rate × consumption per head × return rate.
2. Non-domestic and governmental volume = 22 % and 14 % of the domestic volume, the distributed ratios for Adh Dhahirah (G1-p60 Table 11).
3. That volume is placed on the commercial and government plots identified by account tariff. It is not spread across the population, so a residential-only branch carries no commercial flow.
4. G1 Table 12 is not applied. It is priced in pupils, beds, employees and floor area, and none of these have been supplied, so the condition in §7.3.2 is unmet in substance. It is adopted the day real quantities arrive, and never together with the ratios.

Four streams sit outside the ratios and are added separately, because the guideline excludes them by name: identified projects such as the industrial estate, special consumption such as labour camps, potable and

¹ Ibri town alone measures 6.21. Five settlements are unusable pending boundary review: Sayh Al Masarrat, Al Jahli, Al Akheedat, Satwah and Al Qali. The figure is recomputed when the clean plot layer arrives.

sewage tankers, and private wells and other non-network abstraction. Tanker water generates sewage at the same 85 % return rate as domestic supply.¹

Table 14 Rules within the load basis

Rule	Statement	Basis
Where the load goes	Every plot, built, planned and unparceled buildings alike, carries its full saturation load in the pipe and civil design, accumulated with the peak factor and with no timing. Dated years use zone totals only: the projection capped at the zone ceiling, with the surplus spilled to adjacent zones in proportion to their vacancy. The two meet only at trunk nodes	Project doctrine, 15 August 2026
Phased elements	Treatment trains, pumps and force-main duty equipment are phased on the dated years; buried civil works are built for saturation	Project doctrine
Agricultural plots	An agricultural meter is an irrigation pump and generates no wastewater. On 172 of 319 agricultural plots Nama bills a household tariff alongside the farm tariff, so dwelling and pump are metered separately. An agricultural meter on a plot that already carries a domestic meter adds nothing; an agricultural-only plot carries no dwelling	Project decision, 30 August 2026, from the account data
Cost-reflective tariff accounts	499 accounts on the cost-reflective tariff are a consumption class, not a land use. They are the largest consumers and cannot be classified by tariff alone; they are resolved against the plot layer or by site check before any ratio is applied	Open item
Direction of accuracy	One population figure cannot serve every use. Pipe sizing and plant staging fail on an under-estimate. Self-cleansing and the early cleansing schedule fail on an over-estimate: a pipe assumed to scour itself while it silts. Capacity is sized on the high case, the cleansing schedule on the low case, and the connection ratio of 0.51 in 2024 rising to 0.61 by 2028 is carried explicitly	G1-p72 §7.4.4 and the Inception Report

12.4 Water demand

Table 15 Water demand criteria (G1 §7.3, pages 59 to 62)

Item	Guideline requirement or value	Reference	Adopted in W13
Domestic consumption, Adh Dhahirah	164 L per head per day. Table 11 values are "indicative figures derived from the recent Integrated Master Plan (2024)", apply "in absence of any updated figures", and "should be validated by NWS as essential design criteria, before designing the project"	G1-p59 to 60 Table 11	164 L/head/day
What the figure measures	Network-accounted domestic water ÷ (occupancy × active domestic accounts). Tanker and private-well supply are not in it	G1-p60	Tanker supply added separately (Section 12.5)
Non-domestic, project-specific method	G1 Table 12 unit rates: education 130 L/day per pupil and staff; hospitals 650 L/day per bed and staff; shopping 12.2 L/day per m ² ; hotels 200 to 500 L per guest per day; offices 93 L/day per employee; restaurants 7.4 L/day per m ² ; mosques 185 L/day per m ² ; dry industry 93 L/day per employee; army camps 185 L/day per occupant; prisons 185 L/day per prisoner and staff; wet industry not applicable, the developer supplies it	G1-p61 Table 12	Not applied until the quantities are supplied (Section 12.3)
G1 Table 12 unit basis	Keyed to floor area, pupils, beds and employees, never to plot area. Applied to cadastral plot area the mosque rate alone gives about four times the whole ultimate plant flow	G1-p61	Plot area is never substituted

¹ G1-p59 §7.3.1: the ratios "do not apply to the water consumption of specific identified non-domestic projects such as economic zones". G1-p61 §7.3.4: special consumption "must be provided by the developer ... These needs are not covered by population forecasts". G1-p70 §7.4 and §7.4.1: other water sources, binding. G1-p71 Table 19: tanker return rate.

Item	Guideline requirement or value	Reference	Adopted in W13
G1 Table 12 deviations	"The designer can provide additional details per category or data for his design and shall substantiate the use of an alternative value"	G1-p61 note	
Non-domestic, planning method	22 % of domestic consumption; the column is headed "Distributed Non-Domestic Ratio (% LPCD)", a governorate water-balance volume spread over population. The ratio can be updated by the NWS planning department	G1-p60 §7.3.2	22 %, as the volume in Section 12.3
Governmental	Project-specific: "not as a ratio of domestic consumption". Planning fallback 14 %	G1-p60 to 61 §7.3.3	14 %, as the volume in Section 12.3
Special consumption	Labour camps and high-water industry must be provided by the developer; not covered by population forecasts	G1-p61 §7.3.4	Additive, outside the ratios
Potable tanker demand	Where a tanker filling station is in or near the project area its consumption must be assessed from NWS filling-station data and added. Table 13: Adh Dhahirah 5,145 m ³ /day in 2021 to 2023, which is 333 % of the 2023 network domestic consumption, the highest in Oman. The master plan holds tanker demand constant	G1-p61 to 62 §7.3.5 Table 13	Tier 1 data request
Double counting	G1 Table 12 or the ratios, never both	G1-p60 to 61	Ratios only, until G1 Table 12 quantities arrive

12.5 Wastewater generation

Table 16 Wastewater generation (G1 §7.4, pages 70 and 71)

Item	Guideline requirement or value	Reference	Adopted in W13
Return rate	Table 19: domestic and tanker 85 % ; non-domestic, government and commercial, 54 %	G1-p70 to 71 §7.4.1	0.85 domestic
Project-specific design flows	To BS EN 752 or an equivalent standard, stated in the calculations, supported by site-specific evidence; the British Water Code of Practice for Flows and Loads is named as a guide	G1-p71 §7.4.1	
Other water sources	"the Designer shall carry out an assessment of other potential water sources within the project area, such as private wells, private water providers, or other non-network abstractions"; "A specific attention is to be taken for catchments covered with private wells"	G1-p70 §7.4, §7.4.1	Tier 1 data request; with tanker water at 333 % of network consumption the network demand alone under-predicts the load
Components	Domestic, non-domestic, tanker discharges and special facilities	G1-p70 §7.4	As guideline

12.6 Peak factors, infiltration and tankers

Table 17 Peaking and allowances (G1 §7.4.2 to §7.4.5, pages 71 to 73)

Item	Guideline requirement or value	Reference	Adopted in W13
Peak factor, primary	Merrimack, Equation 5, "is to be used" for a catchment "having over 100 properties"; peak factor = $Q_{pdf} \div Q_{adf}$	G1-p71 §7.4.2	Merrimack above 100 properties; below 100 the guideline gives no formula and the 100-property factor is held
Peak factor, alternative	Peltier, Equation 6, "the Average Daily Flow in this formula is in liters per second"	G1-p72	Reported as a comparison; no decision rests on it
Peak factor cap	"It is recommended that the hourly peak factor should not exceed 5.0", a recommendation rather than a limit	G1-p72 note	Reported above 5.0, never truncated

Item	Guideline requirement or value	Reference	Adopted in W13
Infiltration	New networks 720 L/day per km of sewer ; existing inland networks 10 % of the wastewater flow; existing networks in groundwater or coastal zones up to 40 %. "Infiltration due to storm water is not considered."	G1-p72 §7.4.3	720 L/day/km assigned to each pipe by length, without a diurnal pattern; the order of peaking and infiltration is to be confirmed with NWS (Section 20)
Infiltration exemption	"Tanker or vacuum collection do not require to account for infiltration volume"	G1-p73	Tanker flow carries none
Sewage tankers	"In 2024, the yellow tanker represented approximately 17 % of the total flow reaching the STPs of Nama WS", a company-wide observation. Collection capacity assumes the same coverage as the water supply, reaching 100 % by the end of the planning period; self-cleansing checked at initial operation. Tanker loads are higher than network effluent	G1-p73 §7.4.4	17 % is not used as a design allowance; the tanker stream is assessed from filling-station data
Treatment plant margin	10 % "when designing new STPs", applied "over and above any redundancies in the design"	G1-p73 §7.4.5; G203-p65 Table 29	

$$Q_{pdf} = 2.65 \cdot Q_{adf}^{0.879} \quad (5)$$

Symbol	Meaning	Unit
Qpdf	Peak daily flow	ML/day
Qadf	Average daily flow	ML/day

$$PF_{WW} = 1.5 + \frac{1}{\sqrt{Q_m}} \quad (6)$$

Symbol	Meaning	Unit
PF _{WW}	Wastewater peak factor, Peltier form from the 2024 Integrated Master Plan	–
Q _m	Average daily flow	L/s

12.7 Values applied in the W13 pipeline

The pipeline sizes every pipe on the saturation load of the plots that drain to it. The per-plot flow is Equation 7.

$$Q_{adf,plot} = \frac{OR \times n_{prop} \times q_{ww}}{1000} \quad (7)$$

Symbol	Meaning	Unit
Q _{adf,plot}	Average dry-weather flow of one plot at saturation	m ³ /day
OR	Occupancy rate, people per property	–
n _{prop}	Properties on the plot, counted from electricity accounts	–
q _{ww}	Wastewater generated per head per day	L/head/day

Table 18 Flow values carried by the W13 pipeline

Quantity	Value	Source
Domestic water consumption	164 L/head/day	G1-p59 to 60 Table 11
Domestic return rate	0.85	G1-p70 to 71 Table 19
Wastewater per head, area average	171.3 L/head/day	164 × 0.85 with the 22 % and 14 % uplifts; a residential-only branch generates 139.4 L/head/day

Quantity	Value	Source
Occupancy rate in the test-area run	5.0	Inherited value; project basis is 5.32 (Section 12.2)
Properties per plot	Counted from 33,970 electricity accounts; 1.0 only where a plot has no account	Project data, 19 August 2026
Infiltration	720 L/day per km, unpeaked	G1-p72 §7.4.3
Peak factor	Merrimack; held at the 100-property value below 100 properties	G1-p71 §7.4.2
Peak factor above 5.0	Reported, never truncated	G1-p72 note
Test-area result	Qadf 3,620 m ³ /day; peak 96 L/s	W13 pipeline, 7 September 2026

13 Design life, horizons and climate

Table 19 Design life and horizons (G1 §7.1 and §7.5; scope)

Item	Requirement or value	Reference
Planning life	25 years , which "corresponds both to the ultimate design capacity of the project, as well as the period over which the NPV will be calculated for ... comparing schemes"	G1-p57 §7.1
Asset lifetimes	Civil and structures 50 years; mechanical 20; electrical 15 to 50; instrumentation, control and automation 15; pipework 50. "These asset lifetimes are used for financial asset depreciation calculations"; technical life may differ	G1-p57 Table 10
Scope horizon	Completion plus 25 years, or ultimate saturation; model years start, 2030, 2055 and ultimate; projections at five-year intervals	Scope p3, p14 to 15
Saturation	No more developable plots. Zone ceilings bite inside the design horizon; the whole boundary saturates around 2062 to 2070	Project doctrine
Climatic design, inland	Peak shade temperature 55 °C; maximum daily average 50 °C; maximum yearly average 35 °C; metal in sun 85 °C; relative humidity to 100 %. "All equipment shall be rated for continuous operation under [these] ambient conditions ... and performance guarantees shall be given at these conditions"	G1-p78 to 79 §7.5 Table 24
Climate resilience	Assessed at a 50-year horizon even where the asset life is shorter, with an NWS-agreed climate model at both +2 °C and +4 °C scenarios, including changed flood frequency and intensity in site selection	G1-p33 §4.4

14 Treated effluent

Table 20 Treated effluent criteria (G1 §7.4.6 and §7.4.7, pages 73 to 78)

Item	Requirement or value	Reference
Production ratio	95 % of the plant inflow	G1-p73
Network loss	"For design purposes, a system loss of 10 percent of all produced TSE shall be assumed", so the deliverable volume is about 85.5 % of the plant inflow, not 95 %	G1-p76 §7.4.6.3(b)
Sizing basis	"The TSE system shall be sized to accommodate the peak demand experienced during the summer months." Seasonal factors as a share of summer: December to February 50 %; March to May and September to November 75 %; June to August 100 %	G1-p76 §7.4.6.4 Table 23
Irrigation rates	Summer planting rates: shrubs 20 to 40, palms 120 to 165, other trees 40 to 80 L per plant per day; hedges 10 L per m per day; ground cover and seasonal flowers 10 and grass 12 L per m ² per day; densities of 15 m for trees and 3 m for shrubs. Roads and junctions with mixed vegetation and no data: 10 L per m ² per day, subject to Municipality approval	G1-p73 to 75 §7.4.6.2 Tables 21 and 22
Large consumers	"Any consumer with an average daily demand larger than 500 m ³ /day is to be studied individually", with an irrigation pattern reflecting its actual settings; consumers with storage spread their demand over the storage hours	G1-p76 §7.4.6.4
Sludge production	0.25 kg per m ³ treated, a master-plan baseline described as "a general guideline" and process-dependent	G1-p78 §7.4.7

15 Sewage treatment plant

Table 21 Treatment plant criteria (G203 §10, pages 63 to 74; G1 §4 and §6)

Item	Requirement or value	Reference
Size category	Small under 500; medium 500 to 20,000; large 20,000 m ³ /day and above. The ultimate Ibri flow places the plant in the large category	G203-p65
Design horizon	At least 15 years, projected population plus industrial load	G203-p65
Capacity basis	Average per-head consumption per the latest NWS Integrated Master Plan; a more realistic figure only as an NWS-accepted, designer-justified exception	G203-p65 §10.2.2.1
Site selection	Table 27: accessibility, land and phasing, winds, geology and hydrology, topography to limit pumping, groundwater and flood protection at the 25 and 100 year levels, buffer from residential areas, life-cycle cost	G203-p63 to 64
Flood criteria	25 and 100 year flood levels; the plant fully operational during floods	G203-p63 Table 27(i)
Land footprint	Per m ³ /day: MBR 0.45 to 0.9 m ² ; MBBR 0.9 to 1.8; SBR 0.9 to 1.8; IFAS 1.2 to 2.5; conventional and extended aeration 1.8 to 3.6; constructed wetland over 10 m ²	G203-p64 Table 28
Built footprint	Total footprint of all structures at most 35 % of the allocated land; boundary setbacks at least 5 m	G1-p50 §6.4.2
Buffer zones	Small and medium plants 500 m to residential and industrial; large plants 300 to 1,000 m from odour modelling on the 5 odour unit contour; small plants under 150 population equivalent 10 to 30 m; sewage pumping stations 30 m residential and 20 m industrial. Subject to NWS approval and the environmental and social impact assessment	G1-p43 to 44 §6.1.3.3 Table 8
Incoming flow	Average sewage flow plus infiltration plus a 10 % operational allowance. Hydraulic pass-through sized on the peak hourly flow; biology on the annual average flow and load. "The hydraulic and load design shall include recycled liquors and the received tanker volumes if applicable"; a design peak instantaneous flow is also defined	G203-p65 to 66 Table 29, §10.2.2.1
Process lines	"Treatment plants shall be designed using several process lines allowing additional capacity for different design horizons or phases"	G1-p53 §6.6.2
Redundancy	Modular N+1 at site level: N elements meet average-day flows, N+1 meets peak capacity	G1-p33 §4.3
Organic load	At least 60 g BOD ₅ and 80 g TSS per head per day unless justified; COD to BOD 1.8 to 2.2 for domestic sewage	G203-p74
Existing works	For the upgrade or expansion of an existing plant the organic design rests on the measured strength from the client's laboratory records, with a growth increment	G203-p74 §10.3.1
Raw sewage characteristics	Primarily from NWS laboratory data for the catchment. Table 30 defaults, usable only by NWS agreement: BOD ₅ 350 to 400, COD 700 to 900, TSS 400 to 500, TKN 60 to 80, ammonia nitrogen 40 to 50, total phosphorus 10 to 15 mg/L, 20 to 35 °C. Table 31 tanker sewage: BOD 350 to 1,050, COD 1,350 to 5,000, TSS 900 to 4,300 mg/L	G203-p67 Table 30; p68 Table 31
High-strength influent	Above a COD to BOD ratio of 2.2 the influent is "not effectively biologically treatable" and a separate treatment line is required where the volume is significant; the tanker stream triggers this	G203-p74 §10.3.1
Organic peak factors	BOD and COD 1.5 for a small plant and 1.2 for medium and large; TKN 2.0 and 1.5; equalisation considered where surge loads are critical	G203-p74 §10.3.1
Tanker reception	A dedicated tanker discharge station with screening and oil and grease removal; a flow-equalisation tank specifically for tanker discharge; the 48 to 72 hour emergency lagoon where NWS requires it; pre-acceptance sampling, procedures, traceability and contractual limits. Pre-treatment selected for tanker and non-domestic sewage	G203-p73 §10.3.1; G1-p55 §6.8
Emergency lagoon	48 to 72 hours of plant capacity "in specific cases, under NWS requirement and approval", for holding problematic influent or effluent and returning it to the inlet works; distinct from the scope's five-day storage lagoons	G203-p73; scope p13(61)

Item	Requirement or value	Reference
Effluent quality	"At least Class A" under MD 145/93 for wadi discharge, subject to prior agreement with NWS and to Environmental Authority and APSR approval. The MD 159/2005 nitrogen and phosphorus limits apply only "if the Wadi discharges to sea at a reasonable distance"; Ibri's wadis are inland	G203-p72 to 73 §10.2.4.3
Nitrogen and chlorine	Total nitrogen below 15 mg/L as N; chlorine residual at the plant outlet 0.3 to 1.0 mg/L, up to 3.0 mg/L permitted to allow decay along the treated effluent network	G203-p71 §10.2.4.1
Nature-based treatment	Constructed wetlands are limited to works of about 500 m ³ /day	G203-p101

16 Water and treated effluent networks

Table 22 Network criteria from the water and TSE guideline (G2 §7 and §9)

Item	Requirement or value	Reference
Transmission velocity	1.0 m/s or more and below 2.0 m/s; 1.5 m/s is the common practical peak over a 25-year horizon	G2-p103 to 104
Head loss equations	Darcy-Weisbach for large pipes and high velocity, Hazen-Williams for small diameters; any other needs NWS hydraulic team approval	G2-p104
Roughness with age	Ductile iron C 140 falling to 120 and ϵ 0.26 rising to 0.45 mm over 20 years; GRP C 150, ϵ 0.005 mm; HDPE C 150, ϵ 0.007 mm; PVC C 150, ϵ 0.0015 mm	G2-p104 Table 21
Treated effluent penalty	Roughness 30 % higher and Hazen-Williams C 10 % lower than the potable values	G2-p104 note
Maximum linear head loss	Transmission below 5.0 m/km; distribution below 3.0 m/km	G2-p105, p136
Distribution velocity	0.4 m/s or more and below 1.5 m/s; below 0.4 m/s a water-quality or age model and proof of a 0.2 mg/L chlorine residual	G2-p136
Distribution pressure	At least 1.5 bar at the worst point at peak hour; at most 4 bar; pressure stays positive under fire flow	G2-p137
Distribution pipe material	PE100 up to 1,000 mm in the fixed outside-diameter series; ductile iron above 300 mm and at road and wadi crossings	G2-p138
Pumping, storage and surge	Storage classes in §5, pumping station design with NPSH and transients in §6, surge analysis mandatory per §10	G2-p53 onward, p70 onward, p144
Tanker filling stations	Peak-hour factor 1.5 to 2.0 on average flow; at least two concurrent bays; 20 % reserve; at least 1 m ³ per minute per bay at 2 to 4 bar	G2-p154

17 Odour and hydrogen sulphide

Table 23 Odour and hydrogen sulphide criteria (G203 §11, pages 162 to 191)

Item	Requirement or value	Reference
Designer deliverable	Designers of sewage networks, pumping stations and force mains shall provide NWS a dedicated hydrogen sulphide management evaluation, preventive (re-sizing, chemical injection) and corrective (monitoring, flushing, coatings, odour treatment)	G203-p162 §11.1
Network design for odour	Retention times minimised, turbulence avoided, no over-sizing of trunk sewers, which creates deposition	G203-p166 §11.3.2; p168 §11.4.2
Pumping and force mains	Avoided wherever gravity is feasible and cost-effective; hydraulic detention minimised, with twinned force mains considered at low flow; force-main discharges submerged where possible; the main allowed to drain back to the wet well between cycles; a partially-full main avoided	G203-p181 to 182 §11.5.3.1
Risk assessment	Qualitative Fayoux scoring on temperature, residence time, velocity and redox (0 to 5 no risk, 5 to 10 low, 10 to 30 significant or certain), evaluated at maximum temperature and night-time flow. Quantitative Pomeroy-Parkhurst, with effective $BOD = BOD_5 \times 1.07^{(T - 20)}$	G203-p185 Table 99; p186 §11.5.3.4
Odour limits at the boundary	Urban or near residential 3 to 5 odour units per m^3 ; industrial 10 to 15; rural or isolated 10 to 20. Compliance at the boundary wall: 5 odour units per m^3 at sensitive receptors, 15 at industrial, with at least three continuous online electronic noses. MD 41/2017 and BS EN 13725:2022	G203-p165 §11.3.1; p170 Table 90
Odour treatment	Abatement assessment and dispersion modelling to justify the odour unit or its absence; in most cases multi-stage, bio-trickling filter plus chemical scrubber with carbon where needed, removing 99.95 % of hydrogen sulphide and 85 % of odour; N+1 redundancy on every odour control system	G203-p170 to 176 §11.5.2
Network emergency storage	Retention basins sized for 24 hours of nominal flow, emptied by sewage tanker; relief sewers at 1.5 times nominal flow. Distinct from the plant's 48 to 72 hour emergency lagoon	G203-p191 §12.1.1
Trade effluent	Non-domestic discharges to sewer comply with Appendix 3 of Royal Decree 115/2001, under a customer agreement with NWS and with NWS-validated pre-treatment where needed	G203-p20 §3.10

18 Modelling, options appraisal and project workflow

Table 24 Modelling and appraisal criteria (G1 §1.6, §11, §12 and §13; Appendix III)

Item	Requirement or value	Reference
Hydraulic model method	To the WaPUG/CIWEM Code of Practice and the US EPA SWMM manuals; modelled at design and again after construction; static, extended-period and surge models submitted to NWS after each design phase, including the final as-built model	G1-p144; p109 §13.4.2
Calibration acceptance	Peak flow within 10 to 15 %; volume within 15 %; correct peak-arrival timing; pump run time within 10 %	G1-p145 Table 32
Options appraisal	At least three options with equivalent function and reliability: one sustainability-led, one international best practice, one established local practice. Evaluated over 25 years at a 5 % discount rate unless NWS instructs otherwise; NWS sets the multi-criteria weights; options within 10 % on total cost are decided on sustainability	G1-p95 to 96 §12; p99; p106
Options per system	Three for the sewer network, three for the treated effluent network, three for each treatment plant. Lifting stations are not a separate set: their number follows from each network layout. Seven criteria: total lifetime cost; sustainability; social development and in-country value; adaptability and resilience; operability; constructability; environmental impact. Sensitivity on the weighting, the discount rate and the input criteria	Project doctrine, 31 August 2026, on G1-p95 to 106
Financial method	Net present value and life-cycle cost at 5 % over 25 years decide; payback is reported but does not decide. The guidelines state the requirement and not the method, so ISO 15686-5 is cited. Operating cost is built bottom-up from duty, never as a percentage of capital; gravity sewers are priced by diameter and depth band; every option, including a costed do-nothing baseline, carries its own pumping, storage and energy	Project doctrine, 1 September 2026, on G1-p57 §7.1 and p95 to 96
Value engineering	A formal study by an independent certified consultant at concept and preliminary stages for projects of OMR 5 million or more; the footnote lowers the trigger to any treatment plant or pumping station over OMR 2 million	G1-p93 §11.2 Table 27
Cost and schedule accuracy	Feasibility ±30 %; preliminary and concept ±20 %; detailed ±10 %. A flood protection assessment at all three phases	G1-p17 to 20 §1.6 Table 2
Remote areas	About 25 km or more from a centralised network, or under 500 residents or 100 plots at the end of the design period: septic tanks to the Oman Private Sewage Disposal Code, holding tanks emptied by vacuum tanker, or package plants for 50 to 5,000 inhabitants with at least one day of effluent storage	G1-p80 §8.1; p83 to 84 §8.4

19 Surveys and investigations

Table 25 Survey criteria (G203 §13, pages 197 to 199; G1 §5)

Item	Requirement or value	Reference
Topographic survey	Along the proposed routes with X, Y and Z on the Omani national datum in metric units; existing utilities and adjacent roads with cross-sections near the pipelines; grid, permanent benchmark and invert levels of existing drains on the maps; the designer selects the terrain model	G203-p197
Accuracy by stage	At 95 % confidence: feasibility 1 to 5 m horizontal and 0.5 to 2 m vertical, an existing DEM acceptable; preliminary and concept 0.25 to 1.0 m horizontal and 0.05 to 0.5 m vertical, by UAV photogrammetry with ground control, LiDAR or GNSS RTK; detailed 0.02 to 0.10 m horizontal and 0.01 to 0.05 m vertical. The 0.5 m terrain blend used for design is checked against this at the survey	G1-p36 Table 5
Control	Benchmarks to ± 0.05 m, at least Order C Class 3 GPS	G203-p198 §13.3
Underground services	Detection covering oil, gas, water and effluent pipelines, cables and irrigation falaj, by trial pits, probes between manholes, ground radar and electro-location	G203-p198 §13.3
Existing assets	CCTV to NF EN 13508-2, a designer obligation to carry out and to provide with the tender documents; CCTV, topographic and geotechnical surveys at the early design stage	G203-p197 to 198
Plant and station sites	Full plot dimensions, GPS coordinates of every boundary corner, surface strata, coverage extending 10 m beyond the plot limits	G203-p198 §13.2
Geotechnical	Boreholes to 5.0 m below the pipe invert for shallow pipelines. Secondary and tertiary gravity sewers: a trial trench to 3.0 m or a borehole beyond, at 100 m spacing; primary sewers and force mains at 500 m. Geophysical investigation at the earliest stage for plants, stations and major wadi, falaj and road crossings	G203-p199 §13.4; G1-p40 to 41 Table 7

20 Register of project decisions and assumptions

Every value below is ours rather than the guideline's. A decision stands until the engineer changes it. An assumption is pending data and is replaced when the data arrives; each states what replaces it.

Table 26 Decisions and assumptions carried by the design

Item	Value	Status	Basis and what replaces it
Maximum cover	12.0 m, no exception, at chambers and along the trench	Decision, 7 September 2026	The guideline's trigger is excavation cost; no costing exists at concept. Revisited per excursion when the cost analysis exists
Tractive tension	1 Pa	Assumption, gap 9	G203 gives no numeric value. Confirm with NWS; at 2 Pa a large share of DN200 pipes need a steeper gradient
Minimum tractive design flow	1.5 L/s	Assumption	Mara's minimum design flow; without a floor the equation demands unbounded slopes as the flow tends to zero
PVC-U wall class	SDR34, SN8	Assumption	Sets the true bore behind the hydraulics; the actual class per PAM-SPC-207 from NWS
Occupancy rate	5.32 project basis; 5.0 in the W13 test-area run	Decision, 30 August 2026; open item for W13	Measured from settlement population and counted domestic properties; recomputed when the clean plot layer lands. The test-area gate figures were produced at 5.0 and the full-area run is to apply 5.32
Properties per plot	Counted; 1.0 fallback	Data, 19 August 2026	Electricity accounts; the fallback applies only to a plot with no account
Wadi ground	Flood hazard classes 4, 5 and 6 of the 50-year grid	Assumption	Stands in for the guideline's washout criterion; settled by a scour-depth check
Reduced cover at a wadi crossing	Not available	Position	The guideline is silent; the 1.5 m answers scour and the encasement answers load
Lateral length cap	45 m on riders and lateral sewers alike	Decision	The conservative reading of G203 Table 6 and §3.2
Inlet angle	85° applied; sharper than 75° flagged	Decision, 20 August 2026; deviation	Avoids about 200 bend chambers that served no construction purpose; see Section 21
Gradient steps	0.05 %	Decision, 23 August 2026	The drawing value is the design value
Chamber spacing	100 m split, rounded to 10 m, 5 m fallback	Decision, 18 August 2026	Satisfies every spacing class of G203 Table 12
Chamber separation	3.0 m	Convention	No minimum exists in the guidelines
Chamber sizes	DN1000 to 3 m, DN1200 to 6 m, DN1500 deeper	Convention	The guideline says only sufficient size
Kerb clearances for gravity sewers	1 m pipe, 0.5 m chamber	Inference	A force-main clause at G203-p51 applied to gravity
House drain outlet	0.60 m below plot ground; 2 % blended fall	Method choice	Concept-stage connectability test only
Wall and bedding allowance	0.05 m	Assumption	Below the crown cover in the invert-depth calculation
Infiltration order	Added unpeaked	Assumption	G1 does not state whether infiltration is peaked; confirm at kickoff
Peak factor below 100 properties	Held at the 100-property value	Method choice	The guideline prescribes no formula below 100
G1 Table 12 drivers	Not applied	Decision, 30 August 2026	Floor area, pupils, beds and employees are not supplied; adopt G1 Table 12 when real quantities arrive, never with the ratios
Road corridor thresholds	Collinear 10°, bend 30°, gate search 45 m, frontage 40 m	Method choices	Tuned on review

Item	Value	Status	Basis and what replaces it
Dual carriageway rules	Excluded; crossings ≤ 70 m, within 25° of square, charged 2,500 m; underpass radius 30 m	Decision, 19 to 21 August 2026	Section 9
Join charge onto the main pipe	400 m equivalent	Decision, 20 August 2026	Fewer than 15 joins costs a pumping station on the test area
Pumping pockets	Under 50 plots absorbed	Rule	Detail design item
Cascading stations; rising-main duty rule	Not settled	Open, 7 September 2026	Decided when the design raises them

21 Deviations to be declared to NWS

Each item below departs from the letter of a guideline clause. Each is stated in the deliverable and NWS agreement is sought.

Table 27 Deviations and the agreement sought

Deviation	Guideline clause	What the design does and why
Inlet angle of 85° rather than 90°	G203-p30, "shall"	Enforcing 90° at every street junction inserted about 200 bend chambers a few metres short of junctions for no construction benefit. The design applies 85°, flags any inlet sharper than 75° for a purpose-made chamber with a curved channel, and never adds a chamber to satisfy the rule
Occupancy from electricity accounts	G1-p58 defines the rate as population ÷ NCSI housing units	Housing units are not published at settlement level, so counted domestic properties stand in for them. Validated by coverage within 2.0 points (Section 12.2)
Spatial allocation of non-domestic flow	G1-p59 §7.3.1 describes "spatially distributed non-domestic consumption" added to domestic consumption	The uplift is concentrated on the commercial and government plots rather than spread across the population. The total is preserved; a residential-only branch generates no commercial flow. Measured effect: project total unchanged, zone totals move between –17 % and +127 %
Planning-tier ratios in a project-specific design	G1-p60 §7.3.2 and p61 §7.3.3, "shall" use G1 Table 12 where detailed land use exists	The G1 Table 12 quantities have not been supplied, so the condition is unmet in substance. The 22 % and 14 % ratios set the volume as a labelled fallback and are replaced by G1 Table 12 when the data arrives
Weekly peak of 20 % in the Inception Report	G1 §7.4.2 gives Merrimack or Peltier only	The Inception Report applied a water-side peak-week concept on top of Peltier. The two peak twice; an NWS ruling is sought (Appendix A)
Infiltration at 10 % of flow in the Inception Report	G1-p72 §7.4.3 gives 720 L/day/km for new networks	The design applies 720 L/day/km per pipe; the Inception Report value is flagged for kickoff
Presence in a wadi where no other route exists	G203-p30 "must be avoided"; p33 "shall be avoided"	The design avoids it; where it cannot, the presence is a recorded and justified exception carried in the deliverable

Appendix A Inception Report values against the guideline

The Inception Report Revision 0 and its demand workbook adopted the values below. Each is reconciled against the guideline clause; where the two differ the item is carried to Section 21.

Table 28 Inception Report R0 adoptions reconciled

Item	Inception Report value	Against the guideline
Population basis	NCSI wilayat series, Ibri 183,564 in 2024, growth about 2.4 to 3.0 % a year, settlement disaggregation	Consistent with G1-p58
Domestic consumption, Adh Dhahirah	163.5 L per head per day, computed from actual consumption 2021 to 2024	Agrees with the 164 of G1 Table 11
Return ratios	Domestic and tanker 0.85; non-domestic and governmental 0.54	Equal to G1-p71 Table 19
Infiltration	10 % of wastewater flow, settlement-conditional	G1-p72 gives 720 L/day/km for new networks; flagged at kickoff
Tanker catchment	Settlements within 25 km of the plant	The 25 km is in G1-p80 §8.1, which defines a remote area as "approximately 25 km or more from existing centralized ... networks" or under 500 residents or 100 plots at the end of the design period. The report reused it as a tanker-catchment radius; the reading is confirmed with NWS
Weekly peak	+20 %	Not in the G1 wastewater peaking chain. G1-p62 §7.3.6 defines a water-side peak day as "the average day consumption in the peak week (7-days rolling) excluding leakage"; the report appears to have carried it into the wastewater chain on top of Peltier. An NWS ruling is sought
Plant margin, effluent ratio, sludge	+10 %; treated effluent 95 % of inflow; sludge 0.25 kg/m ³	All three in G1-p73 and G1-p78 §7.4.7
Treated effluent network loss	Not applied	G1-p76 requires a 10 % loss on produced effluent; the 95 % ratio alone overstates the deliverable volume by about 10 %